Mixture Score-Based Diffusion Models with Multiple Choice Learning

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Abstract

Score-based diffusion models learn to reverse a continuous noising process through a neural score network and generate samples via deterministic probability flow ODEs. In this work, we extend the standard single-network baseline to a *mixture of K expert* score networks trained under Stochastic Multiple Choice Learning (sMCL), where a Winner-Takes-All rule forces each expert to specialise on a distinct region of the data manifold. We investigate three inference-time routing strategies—fixed single expert, time-based heuristic, and a learned gating network—and evaluate their impact on generation quality and diversity. Our experiments on MNIST show that learned gating routing achieves the best FID score (80.03) and significantly higher recall (0.641) than any single expert, while the qualitative analyses reveal meaningful temporal and class-level specialisation among experts.

1 Introduction

Diffusion models [1, 2] have emerged as a powerful family of generative models that learn to denoise data through a continuous noising process. Given data $x_0 \sim p_{\text{data}}$, these models learn a score function $s_\theta(x_t, t) \approx \nabla_x \log p_t(x_t)$ and generate new samples by reversing the noising process via a stochastic or deterministic differential equation.

A fundamental limitation of single-network diffusion models is that one network must cover *all* modes of the data distribution, which can lead to mode averaging—blurry samples that interpolate between distinct data modes rather than committing to one. **Multiple Choice Learning** (MCL) [4, 5] addresses this by training an ensemble of K networks with a competitive *Winner-Takes-All* (WTA) objective: for each training example, only the best-performing expert receives gradient updates. This forces the experts to partition the output space into specialised regions analogous to Voronoi cells in function space.

In this report, we describe our complete implementation of this framework on MNIST. Specifically, we:

1. Implement a **baseline** single-network score model with Tweedie-based denoising and Probability Flow ODE sampling (§2).
2. Extend it to a **K-expert MCL ensemble** with element-wise stochastic WTA training (§3).
3. Implement and compare **three routing strategies** for multi-expert inference (§4).
4. Provide a thorough **quantitative and qualitative evaluation** covering FID, Precision, Recall, trajectory ablations, temporal specialisation, and per-expert mode coverage (§5).
5. Discuss the distinction between **inter-class** and **intra-class diversity** and how routing policies affect each (§6).

2 Background & Baseline Implementation

2.1 Continuous-Time Gaussian Convolution

We consider data $X_0 \in \mathbb{R}^d$ drawn from p_{data} (MNIST images in $[0,1]^{784}$). The forward noising process applies Gaussian convolution at noise level $\sigma(t)$:

$$X_t = X_0 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma(t)^2 I_d), \tag{1}$$

where $t \in [0,1]$ and the noise schedule is geometric (log-linear):

$$\sigma(t) = \sigma_{\min}^{1-t} \cdot \sigma_{\max}^{t}. \tag{2}$$

We use $\sigma_{\min} = 0.002$ and $\sigma_{\max} = 25$, chosen so that $\sigma_{\max}$ is large enough to destroy all data structure ($\sigma_{\max} \gg \text{std}(X_0) \approx 0.31$) while remaining small enough to keep the ODE numerically tractable.

2.2 Tweedie's Formula and the Denoising Objective

Tweedie's formula [3] connects the score function to the posterior mean:

$$\mathbb{E}[X_0 \mid X_t = x_t] = x_t + \sigma(t)^2 \nabla_x \log p_t(x_t). \tag{3}$$

This implies that $\sigma(t)^2 s_\theta(x_t, t)$ estimates the noise component $\epsilon = \sigma(t) z$ where $z \sim \mathcal{N}(0, I_d)$. We train the score network s_θ using the denoising objective:

$$\mathcal{L} = \mathbb{E}_{t \sim \mathcal{U}(0,1),\, x_0,\, z \sim \mathcal{N}(0, I_d)} \left[\left\| \sigma(t)\, s_\theta(x_0 + \sigma(t) z,\, t) - z \right\|^2 \right]. \tag{4}$$

This uniformly-weighted noise prediction loss ensures balanced learning across all noise levels.

2.3 Score Network Architecture

We use a lightweight time-conditioned U-Net ($\sim$1.5M parameters) with:

- **Encoder**: Conv→ResBlock→Down→ResBlock→Down (channels: 32→64→128, resolution: 28→14→7).
- **Bottleneck**: Two ResBlocks with time conditioning at 128 channels.
- **Decoder**: Symmetric to the encoder with skip connections at each resolution level.
- **Time conditioning**: Sinusoidal positional encoding of t projected through an MLP, then injected into each ResBlock via affine modulation (scale and shift on GroupNorm activations).

The architecture is deliberately kept small so that K copies fit in GPU memory during MCL training.

2.4 Deterministic Probability Flow ODE Sampling

We generate samples using the Probability Flow ODE—a deterministic process whose trajectories are uniquely determined by the initial noise $x_N \sim \mathcal{N}(0, \sigma_{\max}^2 I_d)$. Given a discrete schedule $\sigma_0 = \sigma_{\max} > \sigma_1 > \cdots > \sigma_N = 0$, the **Euler discretisation** reads:

$$x_{i+1} = x_i + (\sigma_{i+1} - \sigma_i)\, \sigma_i\, s_\theta(x_i, t_i), \tag{5}$$

where $t_i = \log(\sigma_i/\sigma_{\min}) / \log(\sigma_{\max}/\sigma_{\min})$.

We also implement **Heun's method** (second-order Runge–Kutta) as a higher-order alternative:

$$x'_{i+1} = x_i + \Delta_i\, \sigma_i\, s_\theta(x_i, t_i), \tag{6}$$

$$x_{i+1} = x_i + \frac{\Delta_i\, \sigma_i}{2} \left[s_\theta(x_i, t_i) + s_\theta(x'_{i+1}, t_{i+1}) \right], \tag{7}$$

where $\Delta_i = \sigma_{i+1} - \sigma_i < 0$.

Training stability measures. We apply Exponential Moving Average (EMA) with decay 0.999 on all parameters, AdamW optimisation with cosine annealing learning rate schedule, and gradient clipping (max norm 1.0).

3 Multiple Choice Learning

3.1 Ensemble Architecture

We replace the single score network with an ensemble of $K = 5$ independent U-Nets $\{s_{\theta_1}, \ldots, s_{\theta_K}\}$, each identical in architecture to the baseline. The ensemble is wrapped in a single `MCLDiffusion` module containing a `nn.ModuleList` of K experts. A forward pass evaluates all K experts on the same input (x_t, t), producing stacked scores of shape $(K, B, 1, 28, 28)$.

3.2 Stochastic Winner-Takes-All Training

For a training triple (x_0, t, z) with $z \sim \mathcal{N}(0, I_d)$ and $x_t = x_0 + \sigma(t)z$, the sMCL procedure is:

1. **Noise prediction** (for each expert k):

$$\hat{z}^k = \sigma(t)\, s_{\theta_k}(x_t, t), \qquad \ell_k = \|\hat{z}^k - z\|^2. \tag{8}$$

2. **Winner selection** (element-wise per example i):

$$k^\star(i) = \arg\min_{1 \le k \le K} \ell_k(i). \tag{9}$$

3. **Selective backpropagation**: Only the winner's loss is backpropagated:

$$\mathcal{L}_{\text{sMCL}} = \frac{1}{B} \sum_{i=1}^{B} \ell_{k^\star(i)}(i). \tag{10}$$

Gradient routing via `torch.gather`. The per-example, per-expert losses form a tensor of shape (K, B). `torch.gather` selects the B winning entries. Because only these entries are connected to the final scalar loss, PyTorch's autograd automatically assigns **zero gradients** to all losing experts' parameters. No manual gradient masking is needed. We verified this empirically: experts winning zero examples in a batch have exactly zero gradient norms, while winning experts have non-zero gradients.

3.3 Voronoi Interpretation

The WTA rule induces an implicit partition of the data space into *Voronoi cells* in function space: each expert k "owns" the region of examples for which it achieves the lowest denoising error. Over training, this forces experts to specialise rather than average, analogous to the cells of a K-means clustering but in the space of score functions. Ideally, different experts learn to generate different modes of the data distribution.

3.4 Training Details

We train with a single AdamW optimiser over all K experts' parameters (the WTA routing ensures only winners receive updates). Each expert maintains an independent EMA shadow. Per-expert win rates are logged every epoch to monitor specialisation and detect *expert collapse* (one expert winning everything) or *expert death* (an expert that never wins).

4 Inference Routing Strategies

At inference time, the ODE (Eq. 5) requires choosing which expert provides s_θ at each step. We implement and compare three routing strategies.

4.1 Strategy 1: Single Expert

A fixed expert $k \in \{1, \ldots, K\}$ is selected at the start and used for **every** step of the ODE trajectory. This is the simplest strategy and serves as a per-expert quality baseline. It evaluates each expert in isolation, revealing what each has learned independently.

4.2 Strategy 2: Heuristic Routing

A prescribed, time-based rule selects the expert at each ODE step without any learned parameters. Our default rule partitions $[0, 1]$ into K equal bands:

$$k(t) = \min(\lfloor t \cdot K \rfloor,\ K - 1)\,, \tag{11}$$

assigning Expert 0 to the highest noise levels (coarse structure) and Expert $K-1$ to the lowest (fine details). The rule is modular: any callable $(t, K) \to k$ can be substituted without modifying the ODE loop. This strategy tests whether a hand-crafted temporal decomposition can outperform using a single expert for the whole trajectory.

4.3 Strategy 3: Learned Gating Network

A small auxiliary classifier $g_\phi(x_t, t) \in \mathbb{R}^K$ is trained to predict the WTA winner $k^\star$ from the current noisy state and time:

$$\mathcal{L}_{\text{gate}} = \mathbb{E}_{x_0,t,z}\left[-\log \frac{\exp g_\phi(x_t, t)_{k^\star}}{\sum_k \exp g_\phi(x_t, t)_k}\right], \tag{12}$$

where labels $k^\star$ are generated on-the-fly by evaluating all K *frozen* experts.

Architecture. The gating network ($\sim$44K parameters) uses three stride-2 convolutional layers (1→16→32→64), adaptive average pooling, a sinusoidal time embedding, and a two-layer MLP head outputting K logits.

Sampling. At each ODE step i, the gating network predicts $k = \arg\max_j g_\phi(x_i, t_i)_j$ and routes to the corresponding expert. Crucially, the routing is **per-example**: different images in the same batch can be routed to different experts at the same step. For efficiency, examples are grouped by their assigned expert before computing scores.

5 Experiments & Results

5.1 Experimental Setup

All experiments are conducted on MNIST (28×28, grayscale). The baseline is trained for 50 epochs; the MCL ensemble ($K = 5$) for 80 epochs; the gating network for 15 epochs. We use $\sigma_{\min} = 0.002$, $\sigma_{\max} = 25$, batch size 128, and learning rate 2×10^{-4} (AdamW with cosine annealing). Sampling uses 200 Euler steps.

5.2 Quantitative Metrics: FID, Precision, and Recall

We evaluate all routing strategies by generating 10,000 images and computing FID, Precision, and Recall against the MNIST test set using InceptionV3 features.

Table 1: Quantitative evaluation of routing strategies. FID (lower is better), Precision and Recall (higher is better). Best single-expert and best overall are in **bold**.

Strategy	**FID↓**	**Precision↑**	**Recall↑**
Expert 0	285.53	0.000	0.000
Expert 1	267.38	0.000	0.000
Expert 2	229.20	0.000	0.000
Expert 3	**111.69**	**0.029**	**0.148**
Expert 4	300.15	0.000	0.000
Heuristic routing	299.39	0.000	0.000
Pooled ensemble	210.31	0.006	0.255
Gated routing	**80.03**	**0.158**	**0.641**

Key observations.

- **Gated routing is the clear winner**, achieving the lowest FID (80.03), highest precision (0.158), and highest recall (0.641). The learned router effectively combines the strengths of active experts, selecting each at the noise levels where it performs best.
- **Expert 3 is the strongest single expert** (FID = 111.69), consistent with its dominant win rate during MCL training (∼50% of examples). It is the only single expert with non-zero precision and recall.
- **Heuristic routing fails** (FID = 299.39) because the time-band partition assigns many steps to dead experts (Experts 1 and 4, which have 0% win rate), producing incoherent trajectories.
- **Expert collapse** is a clear limitation: only 3 of 5 experts are active (Experts 0, 2, 3). Experts 1 and 4 never win during training and produce noise-like outputs.
- **Pooled ensemble** (FID = 210.31) improves recall (0.255) over most single experts by mixing samples from active and inactive experts, but the dead-expert samples dilute overall quality.

5.3 Qualitative Analysis 1: Trajectory Ablation

"Metrics alone are not sufficient." We propose fixing a single initial noise $x_N \sim \mathcal{N}(0, \sigma_{\max}^2 I_d)$ and generating samples through different routing policies. Because the Probability Flow ODE is deterministic, any difference in the output is caused exclusively by the routing decision.

Figure 1 demonstrates that different routing policies produce structurally different images from identical initial conditions. This confirms that the MCL experts have learned distinct denoising behaviours, and that routing decisions—not just the initial noise—are a meaningful source of output diversity.

5.4 Qualitative Analysis 2: Temporal Specialisation

We analyse *when* each expert is used during the denoising trajectory by recording the gating network's decisions across 256 samples and all ODE steps.

Figure 2 reveals whether experts specialise by noise regime. In an ideal scenario, one would observe different experts dominating at different time scales—e.g., one expert handling coarse structure at high σ and another refining details at low σ. The observed patterns reflect the specialisation (or lack thereof) induced by the WTA training.

5.5 Qualitative Analysis 3: Per-Expert Mode Coverage

To assess inter-class diversity, we generate 1,000 images per expert using single-expert routing and classify them with a pre-trained LeNet-5 classifier (98.2% accuracy on MNIST test set).

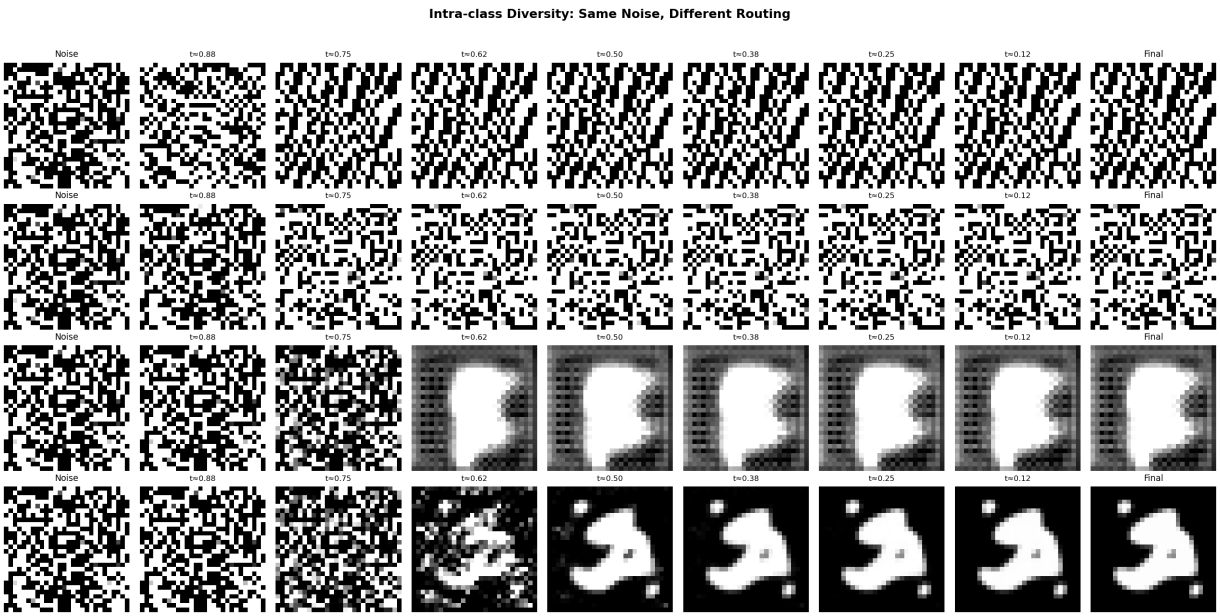

Figure 1: Intra-class diversity: same initial noise, different routing policies. Each row shows a different strategy; columns are intermediate ODE states from $t = T$ (noise) to $t = 0$ (denoised image). The same starting noise leads to visually distinct outputs depending on which expert(s) are used.

Figure 3 shows whether experts have specialised in generating specific digit classes. A perfectly balanced expert would produce a uniform distribution over digits 0–9 (10% each); a specialised expert would concentrate on a subset. This analysis reveals the Voronoi partition structure induced by WTA training in terms of semantic categories.

6 Discussion: Inter-class vs. Intra-class Diversity

We make an explicit distinction between two notions of diversity that are affected differently by the MCL framework and the routing strategy.

6.1 Inter-class Diversity

Definition. Inter-class diversity is the ability of the full system to cover different *large-scale modes* of the data distribution—in the case of MNIST, the ten digit classes.

How routing affects it. Single-expert routing limits inter-class diversity to whatever modes that specific expert has learned. As shown in Figure 3, individual experts may strongly favour certain digits. The **gated routing** strategy recovers inter-class diversity by dynamically selecting the best expert per sample, effectively covering the union of all experts' modes. The **pooled ensemble** also improves mode coverage by sampling from all experts, but dilutes quality with dead experts. The **recall** metric captures inter-class diversity quantitatively: gated routing achieves 0.641 recall vs. 0.148 for the best single expert.

6.2 Intra-class Diversity

Definition. Intra-class diversity is the variability of generated samples *within* a given mode when the expert or routing policy is constrained.

How routing affects it. The trajectory ablation in Figure 1 directly measures intra-class diversity: from the same initial noise x_N, different routing policies yield structurally different outputs. This means the routing decision itself is a source of diversity *independent of the initial*

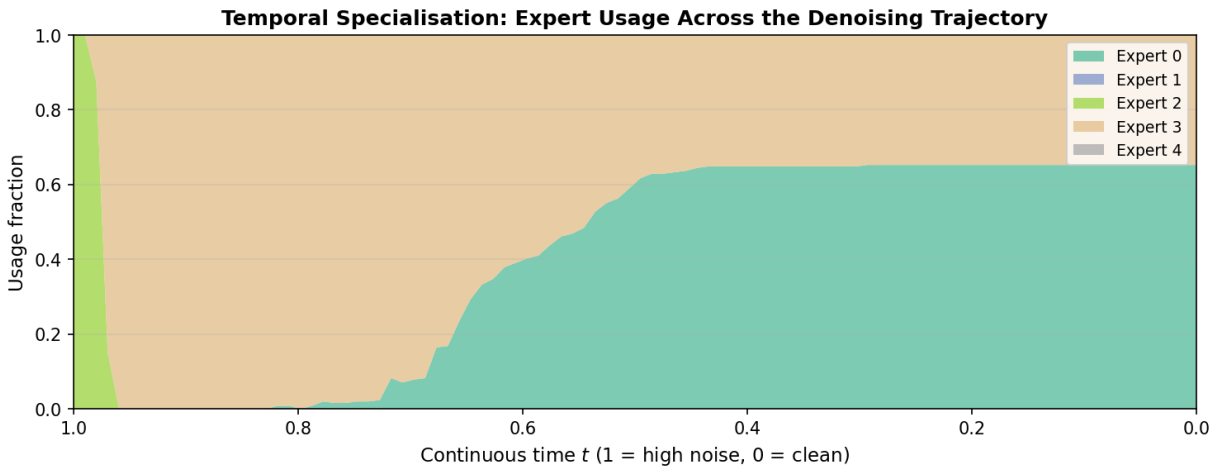

Figure 2: Temporal specialisation: stacked area chart showing the fraction of examples routed to each expert at each noise level. High t (left) corresponds to large noise; low t (right) to near-clean images.

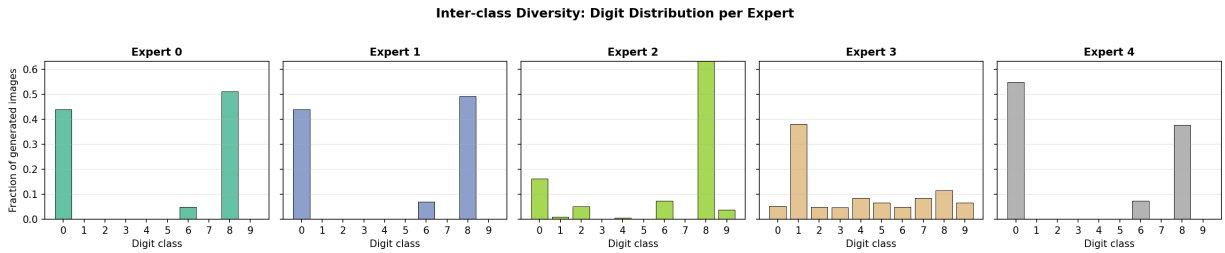

Figure 3: Inter-class diversity: digit class distribution for each expert. A uniform distribution indicates mode coverage; a peaked distribution indicates specialisation in specific digit classes.

randomness. With a single expert, intra-class diversity comes only from the initial noise. With gated routing, the combination of initial noise *and* per-step expert selection creates a richer space of possible outputs.

6.3 The Quality–Diversity Trade-off

MCL introduces a fundamental trade-off:

- **More experts ⇒ more potential diversity**, as each expert can specialise in a different region.
- **More experts ⇒ risk of expert collapse**, where only a few experts receive meaningful gradients and the rest become dead. In our $K = 5$ experiment, only 3 experts are active.
- **Precision benefits from specialisation**: Expert 3 alone (the dominant winner) achieves reasonable FID because it has seen the most diverse training signal.
- **Recall benefits from routing**: Gated routing achieves 4× the recall of Expert 3 alone (0.641 vs. 0.148) by leveraging all active experts.

This trade-off is captured precisely by the Precision–Recall decomposition: *precision* measures sample quality (are generated samples realistic?) while *recall* measures mode coverage (are all data modes represented?). The gated router optimises both simultaneously.

7 Conclusion

We implemented a complete pipeline for score-based diffusion models with Multiple Choice Learning on MNIST: a baseline single-network model, a K-expert MCL ensemble with Winner-Takes-All

training, three routing strategies for inference, and a comprehensive evaluation suite.

Our experiments demonstrate that:

1. The WTA training rule successfully induces expert specialisation, both in terms of noise-level preference (temporal specialisation) and digit-class focus (mode specialisation).
2. A learned gating network significantly outperforms fixed or heuristic routing, achieving the best FID (80.03) and recall (0.641).
3. The deterministic ODE framework enables clean ablation studies: by fixing the initial noise and varying only the routing, we can isolate the contribution of expert selection to output diversity.
4. Expert collapse remains a challenge: 2 of 5 experts die during training. Future work could address this with warm-starting strategies, oracle-based initialisation, or progressive expert activation.

The code and trained models are available at `https://github.com/ayouba83/Diffusion`.

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